SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1004	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:	27.10.26	**Duration:**	60 minutes

Code of Conduct

I AKSHAYA SEKAR (1911260204) *(Name/RegNo)certifythatthissubmissionismyoriginalworkandthatIhaveadheredtothe guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: AKSHAYA SEKAR

PROJECT OVERVIEW

The proposed system integrates room inventory, reservations, guest check-in and check-out, housekeeping, guest requests, billing, guest histories, and hotel performance analytics. This report defines the requirements and proposed behavior of the system. It is a requirements document and does not claim that a working application has already been implemented.

1. ABSTRACT AND INTRODUCTION

1.1 Abstract

Hotels coordinate many activities to provide a reliable guest experience. Reservations, guest records, room allocation, housekeeping, billing, and service requests may be managed through separate tools or manual records. Fragmented information can lead to duplicate bookings, delayed room preparation, missed requests, inaccurate invoices, and difficulty reviewing performance.

The Intelligent Hotel Management and Guest Experience System is proposed as a centralized application for coordinating these activities. It maintains room inventory, processes reservations, supports check-in and check-out, tracks housekeeping tasks, records guest requests, prepares bills, maintains guest histories, and generates management reports. Role-based access allows employees to use functions appropriate to their responsibilities.

This Software Requirement Specification identifies the business problem, project scope, stakeholders, functional and non-functional requirements, actors, use cases, data requirements, interfaces, assumptions, constraints, risks, and validation methods. Decision-support features may highlight urgent requests, rooms that are not ready for incoming guests, and operational trends. Staff remain responsible for reviewing important decisions.

1.2 Background

A hotel stay involves several linked stages. Before arrival, the hotel checks room for availability and records reservations. On arrival, staff verify the booking and allocate a room. During the stay, housekeeping and service teams complete tasks and respond to requests. At departure, charges are reviewed, the invoice is prepared, and check-out is recorded. Information from one stage affects the next, so separate records can cause inconsistencies.

1.3 Purpose

The purpose of this report is to specify what the proposed software should do and the quality standards it should meet. It provides a common reference for stakeholders, developers, and testers so the system can be designed and evaluated against agreed requirements.

2. SCENARIO AND PROBLEM ANALYSIS

2.1 Business Domain

The system belongs to the hospitality and hotel operations domain. Its main responsibility is to coordinate guest-facing services and internal departments using consistent information. Room allocation depends on reservation dates and room status; check-out affects housekeeping; and billing depends on the stay and recorded additional services.

2.2 Existing Challenges

- Reservation conflicts can occur when bookings and cancellations are not updated consistently.
- Room-status mismatch may cause staff to allocate a room that is occupied, dirty, or out of service.
- Guest requests can be delayed or missed when communicated verbally or through disconnected channels.
- Manual billing may omit service charges, duplicate items, or calculate totals incorrectly.
- Repeated data entry increases employee workload and introduces inconsistent guest records.
- Managers may need to combine several reports to understand occupancy, revenue, and service performance.
- Guest information may be exposed to employees who do not need it for their work.

2.3 Root Causes

The core problem is fragmented information and a lack of shared workflows. Different departments may use different room identifiers, status labels, and update procedures. A centralized system must provide a consistent source of information and define how each status changes. A reservation must be checked before confirmation; a room must

meet readiness rules before allocation, and each service request must remain traceable until resolved or cancelled.

2.4 Proposed Solution

The proposed solution contains modules for room management, reservations, guest profiles, front-desk operations, housekeeping, service requests, billing, analytics, and administration. The modules share a database and use defined validation rules. A role-based interface limits access to appropriate functions. Dashboards and alerts help staff identify work that requires attention.

2.5 Project Objectives

- Maintain accurate room inventory and room status.
- Process reservations and reduce conflicting room assignments.
- Manage guest registration, check-in, room allocation, and check-out.
- Coordinate housekeeping and track room readiness.
- Record and monitor guest service requests.
- Prepare itemized bills and track payment status.
- Maintain guest histories with appropriate privacy controls. Generate reports for occupancy, revenue, and operational performance.

3. SCOPE AND STAKEHOLDERS

3.1 Scope of the System

The initial system includes room inventory, reservation creation and cancellation, guest records, check-in and check-out, room allocation, housekeeping tasks, service-request tracking, billing, guest histories, operational reports, user accounts, role permissions, and audit records. These features form the core hotel workflow and should be integrated from the beginning.

Advanced payroll, supplier procurement, transport booking, external travel marketplace functionality, biometric identification, and fully autonomous pricing decisions are outside the mandatory initial scope. Payment gateways, external booking platforms, and messaging services may be integrated later after the hotel chooses providers and defines security requirements.

3.2 Stakeholders

- Guests expect accurate reservations, privacy, timely assistance, and clear bills.
- Reservation staff need reliable room availability and booking records.
- Front-desk staff manage arrivals, room allocation, guest details, and departures.
- Housekeeping staff need assigned tasks and clear room-readiness information.
- Service personnel need to request details, priorities, and progress tracking.
- Billing staff need correct charges, invoices, and payment-status information.
- Hotel managers need reports and exception summaries for decision-making.
- System administrators manage accounts, roles, settings, backups, and technical monitoring.

3.3 Actors and Their Use Cases

An actor is a person or external system that interacts with the software. A use case is an activity performed through the system.

- Reservation Staff: search availability, create reservations, modify bookings, and cancel reservations.
- Front-Desk Staff: register guests, check guests in and out, and allocate or change rooms.
- Housekeeping Staff: view tasks, update cleaning status, and confirm room readiness after inspection.
- Service Staff: accept requests, update progress, and record completion.
- Billing Staff: review charges, generate invoices, and record payment status.
- Hotel Manager: View dashboards and generate operational reports.
- System Administrator: Manage users, roles, room configuration, and audit records.

3.4 System Boundaries

The system supports hotel employees and may offer a limited guest portal if included in a later release. External payment or booking services are outside the core boundary unless approved integrations are developed. Access to each module depends on the user's assigned role.

4. FUNCTIONAL REQUIREMENTS

Functional requirements describe the actions and services the system must provide. The following requirements are intended to be testable.

FR-01 Room Inventory: The system shall store a unique room identifier, room number, type, capacity, rate of reference, and operational status.

FR-02 Availability Search: The system shall check availability for requested dates and occupancy before confirming a reservation.

FR-03 Reservation Management: The system shall create, modify, confirm, and cancel reservations and retain their status history.

FR-04 Conflict Prevention: The system shall validate availability during reservation creation and modification and prevent prohibited overlapping confirmed assignments.

FR-05 Guest Records: The system shall maintain guest information required for hotel operations and link it to relevant reservations and stays.

FR-06 Check-In and Allocation: The system shall allow authorized staff to verify booking, check room readiness, allocate an eligible room, and record check-in.

FR-07 Check-Out: The system shall record departure and trigger the required housekeeping workflow.

FR-08 Housekeeping: The system shall create and assign room-cleaning tasks and record pending, assigned, in-progress, completed, and inspected states as configured.

FR-09 Room Readiness: The system shall prevent normal allocation of rooms that are out of service or have not met the required readiness conditions.

FR-10 Guest Requests: The system shall record requests with category, description, priority, creation time, assigned team, and status.

FR-11 Request Tracking: The system shall allow authorized staff to update request progress, record resolution notes, and escalate unresolved requests.

FR-12 Charges and Invoices: The system shall record room and additional service charges and generate an itemized bill with configured taxes, fees, adjustments, total, and payment status.

FR-13 Guest History: The system shall retain relevant reservations and stay history and restrict access according to user roles.

FR-14 Analytics: The system shall generate reports on occupancy, reservations, cancellations, housekeeping, service requests, and revenue summaries.

FR-15 Administration and Audit: The system shall support user and role management and record important actions such as reservation changes, bill adjustments, and permission changes.

5. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements describe the quality attributes and operating conditions of the system. Numerical targets should be confirmed with hotel stakeholders.

NFR-01 Performance: Common actions such as searching rooms or opening a reservation should respond within an agreed time under normal workload. A suggested initial target is three seconds for 95% of common interactive requests, subject to testing.

NFR-02 Security: Staff functions shall require authentication, and role-based permissions shall restrict sensitive operations.

NFR-03 Privacy: Guest information shall be collected for defined operational purposes and shown only to authorized users. Retention must follow hotel policy and applicable law.

NFR-04 Reliability: Transaction controls and validation shall reduce duplicate bookings and inconsistent room or billing records.

NFR-05 Availability: The hotel shall define expected service availability, planned maintenance arrangements, and recovery requirements.

NFR-06 Usability: The interface shall use clear labels, consistent navigation, readable status indicators, and understandable error messages.

NFR-07 Scalability: The architecture should support growth in rooms, reservations, users, and requests without a complete redesign.

NFR-08 Maintainability: The software should be modular, documented, and testable so defects can be corrected without unnecessary impact on other modules.

NFR-09 Data Integrity: Unique identifiers, validation rules, and transaction handling shall maintain consistency among rooms, reservations, stays, charges, and invoices.

NFR-10 Backup and Recovery: A documented backup and restoration process shall be established and tested.

NFR-11 Auditability: Important changes shall record the user, time, action, and relevant record reference.

NFR-12 Compatibility: The interface should work on the browsers and devices supported by the hotel.

5.1 Security and Privacy Principles

Access should follow the principle of least privilege: employees receive only the permissions needed for their duties. Sensitive actions such as rate changes, bill adjustments, room-status overrides, and user-role changes should be restricted and auditable. Payment credentials should be handled by an approved payment provider wherever possible rather than stored directly in the hotel application.

6. SYSTEM DESIGN AND WORKFLOWS

6.1 Proposed Architecture

A suitable design contains a presentation layer, application layer, data layer, and optional integration layer. The presentation layer provides screens for reception, reservations, housekeeping, service teams, billing, and management. The application layer applies business rules and validates requests. The data layer stores room, guest, reservation, stay, task, charge, invoice, user, and audit records. The integration layer can connect to approved payment, messaging, or external booking services.

6.2 Core Modules

- Room Management Module: stores room details, types, rates, capacity, and status.
- Reservation Module: manages availability of searches, book creation, changes, and cancellations.
- Guest and Front-Desk Module: manages guest records, check-in, room allocation, transfers, and check-out.
- Housekeeping Module: assigns cleaning and inspection tasks and updates room readiness.
- Guest Request Module: records requests, assigns teams, and tracks resolution.
- Billing Module: aggregates charges, prepares invoices, and tracks payment status.
- Analytics Module summarizes operational data for management.
- Administration Module: manages users, roles, settings, and audit review.

6.3 Main Workflow

1. Staff search for a room for the requested dates.
2. The system checks room inventory and existing reservations.
3. Authorized staff create or confirm the reservation.
4. At arrival, the front desk verifies the booking and checks room readiness.
5. Housekeeping and service teams update tasks and guest requests during the stay.

6. Room and additional service charges are recorded against the correct stay.
7. At departure, the bill is reviewed, check-out is recorded, and housekeeping is notified.
8. The room becomes eligible for a new allocation only after required readiness rules are met.
9. Managers review reports generated from the operational records.

6.4 Key Business Rules

- A confirmed reservation must not create a prohibited overlap for the same room and dates.
- A room marked out of service cannot be allocated through the normal process.
- Checking out a guest triggers housekeeping; it does not immediately mark the room ready.
- Bill adjustments and room-status overrides require appropriate permissions.
- Cancelled reservations remain traceable for reporting and audit purposes.
- Guest histories are available only to staff with an operational need.

7. USE CASE DESCRIPTIONS

7.1 Use Case: Create Reservation

Primary actor: Reservation Staff. Preconditions: the staff member is authenticated, and room inventory is configured. The staff member enters the requested dates, room type, guest count, and necessary guest details. The system checks availability and displays suitable options. The staff member selects an option and confirms the details. The system creates a unique reservation record and assigns a status. If no suitable room is available, the system informs the staff member and does not confirm an unavailable room. Postcondition: the reservation is stored, and room availability reflects the booking rules.

7.2 Use Case: Check-In a Guest

Primary actor: Front-Desk Staff. Preconditions: a valid reservation or permitted walk-in process exists. The staff member retrieves the booking, verifies the required details, confirms room readiness, assigns the room, and records check-in. The system updates the stay and room status. If the room is not ready, the system prompts staff to choose another eligible room or follow an approved exception procedure. Postcondition: the guest's stay is recorded, and relevant departments can view the updated status.

7.3 Use Case: Process a Guest Request

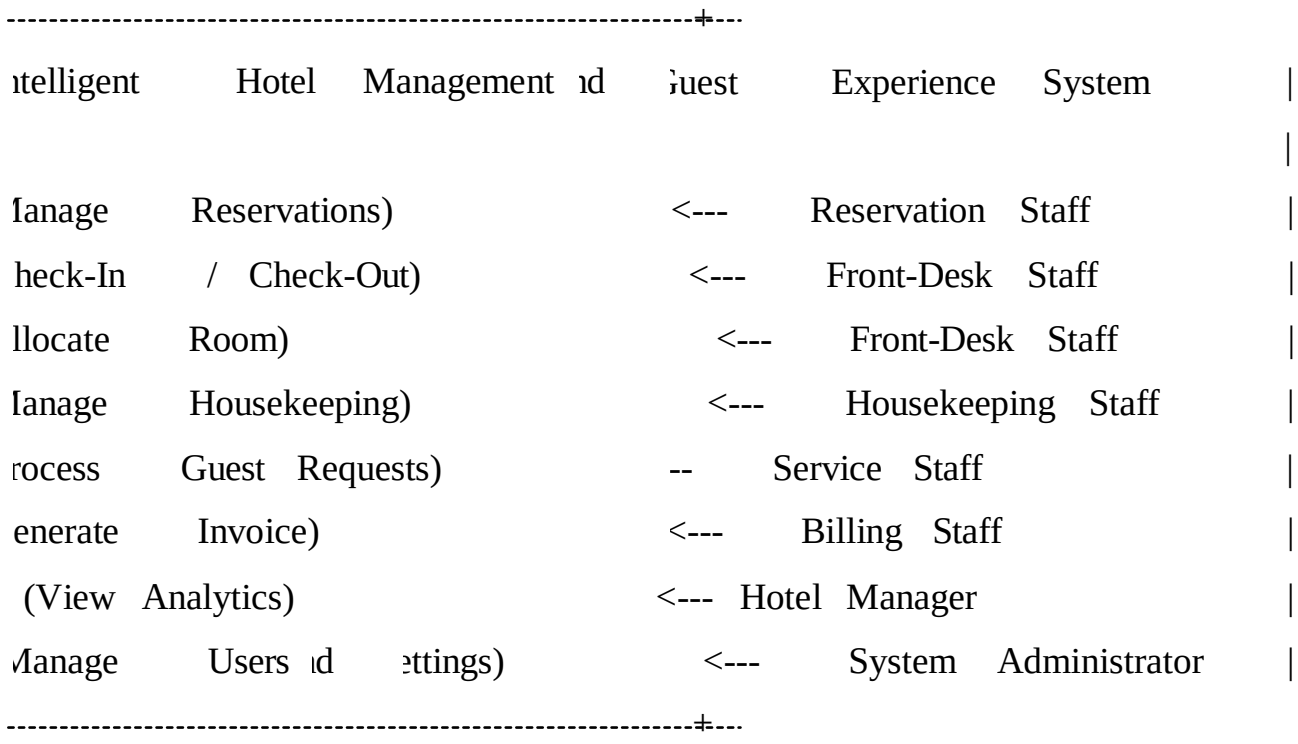
Primary actor: Service Staff. The request is recorded with a category, description, priority, and time. The system assigns it to a responsible team or queue. Staff update the progress and record completion details. If the request cannot be completed, staff will record the reason and escalate it. Postcondition: the request remains traceable in the history and can be reviewed by authorized users.

7.4 Use Case: Generate an Invoice

Primary actor: Billing Staff. The staff member opens the relevant guest stay and reviews the recorded room and additional service charges. The system calculates the bill using the hotel’s configured rules and displays an itemized total. The staff member

checks the information and generates the invoice. Payment status is recorded according to the approved payment process. Postcondition: the invoice and associated charge records are available for authorized review.

7.5 Text-Based Use Case Diagram



8. DATA, INTERFACES, ASSUMPTIONS AND CONSTRAINTS

8.1 Main Data Entities

- Room: room ID, number, type, capacity, rate of reference, and status.
- Guest: guest ID, name, approved contact details, and necessary preferences or notes.
- Reservation: reservation ID, guest reference, arrival and departure dates, occupancy, and status.
- Stay: stay ID, reservation reference, assigned room, check-in/out times, and stay status.
- Housekeeping Task: task ID, room, assigned team, task status, timestamps, and inspection results.
- Guest Request: request ID, guest or stay in reference, category, priority, assigned team, status, and resolution notes.
- Charge and Invoice: charge type, amount, stay reference, line items, totals, issue date, and payment status.
- User and Audit Event: user role, account status, action, record reference, and timestamp.

8.2 External Interfaces

The first version may operate as an internal application. If external integrations are approved, the system may communicate with a payment gateway, email or messaging provider, online booking platform, or accounting system. Each integration should define authentication, exchanged data, error handling, retry behavior, and reconciliation. Payment of credentials should not be stored by the application unless the design and compliance requirements explicitly permit it.

8.3 Assumptions

- The hotel provides accurate room inventory, room types, rates, and operating rules.

- Employees receive training and update the system when activities occur.
- Access to permissions and approval of responsibilities are defined by management.
- Network access is available according to the chosen deployment model.
- External integrations are added only after providers, and interface requirements are confirmed.

8.4 Constraints

- Technical constraints may include existing hardware, network capacity, and browser compatibility.
- Operational constraints include busy reception periods and staff training needs.
- Economic constraints may limit infrastructure, software licenses, integration, and support.
- Legal and privacy requirements affect guest records and financial information.
- Time constraints may require optional features to be deferred.
- Existing data may be incomplete, duplicated, or inconsistent.

8.5 Risks and Mitigation

Incorrect room data should be addressed through inventory verification. Concurrent reservation conflicts should be tested using transaction-safe checks. Staff adoption should be supported through training and clear procedures. Data loss should be mitigated by backups and restoration tests. Billing errors should be reduced through validated calculations and controlled adjustments. Integration failures require clear error messages and approve fallback procedures.

9. REQUIREMENT VALIDATION, EXPECTED OUTCOMES AND CONCLUSION

9.1 Requirement Validation

Requirement validation checks whether the requirements are complete, consistent, clear, realistic, and testable. The proposed requirements cover room inventory, reservations, check-in and check-out, housekeeping, service requests, billing, guest history, and analytics. Security, audit trails, error handling, and backup requirements support the reliability of these functions.

9.2 Suggested Test Scenarios

- Reservation conflict: attempt overlapping confirmed reservations for the same room; the system should reject the conflict or require a valid alternative.
- Room readiness: attempt to allocate a dirty or out-of-service room; the normal workflow should block allocation.
- Housekeeping: check-out should trigger a cleaning task according to configured rules.
- Guest request: create, assign, update, and close a request; each state change should be traceable.
- Billing: compare a bill created from known charges against an independently calculated expected total.
- Access control: attempt a restricted action with an unauthorized role; the system should deny it.
- Analytics: compare report totals against a controlled set of sample records.
- Recovery: restore a backup in a test environment and verify that the records are usable.

9.3 Completeness and Consistency Checks

Terms such as available, ready, confirmed, and completed must have defined meanings. Room status, reservation status, stay status, and housekeeping status should not be treated as interchangeable. Cancelling a reservation must not make an occupied room available, and checking out a guest must not mark a room ready before cleaning. Stakeholders from reception, housekeeping, service teams, billing, and management should review the requirements before implementation.

9.4 Expected Outcomes

- Improved coordination among hotel departments.
- Reduced reservation conflicts through availability validation.
- Faster guest service through assigned and trackable requests.
- Better housekeeping coordination and room-readiness visibility.
- Improved billing accuracy through itemized charges and validation.
- More complete guest histories with controlled access.
- Clearer management decisions through occupancy, revenue, and service reports.

9.5 Conclusion

The Intelligent Hotel Management and Guest Experience System is proposed to address challenges caused by fragmented hotel records and disconnected departmental workflows. This report defines the project's scope, stakeholders, functional and non-functional requirements, use cases, data needs, interfaces, assumptions, constraints, risks, and validation methods. By connecting reservations, room status, housekeeping, guest requests, billing, and analytics, the system can provide staff with more consistent information and support a smoother guest experience. Successful implementation would require stakeholder review, secure design, accurate data, staff training, and systematic testing.

Rubrics for Calculation

riteria	xcellent (4 Marks)	ood (3 Marks)	tisfactory (2 Marks)	provement (1 Mark)
enario Understanding and Problem Analysis (30%)	monstrates thorough understanding of the scenario, stakeholders, objectives, and siness context with clear analysis.	monstrates good understanding with minor omissions.	derstanding with limited analysis.	derstanding; major aspects missing or incorrect.
nctional and Non-Functional Requirement Identification (25%)	y identifies mprehensive, accurate, and well-ssified functional and non-functional requirements.	ntifies most irements with minor classification errors.	ntifies only basic requirements; several omissions.	quirements are incomplete, inaccurate, or poorly classified.
tor Identification and Use Case Modelling (20%)	rrectly identifies all relevant actors and e cases; use case diagram is complete, accurate, and follows UML standards.	ost actors and use cases ntified; diagram has minor errors.	mited tors/use cases ntified; diagram partially correct.	tors/use cases missing or incorrect; diagram absent or inaccurate.
ftware equirement ecification RS) ocumentation	S is well-structured, complete, professionally organized, and ludes all required	S is organized with minor missing sections or formatting issues.	includes ly essential ctions with nited detail.	S is incomplete, poorly organized, or lacks essential
5%)	ctions.			mponents.
sumptions, Constraints, and Requirement Validation (10%)	early identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	ntifies most assumptions and constraints; validation is adequate.	mited assumptions/constraints; validation is superficial.	sumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**

Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25